MEZO BUILDATHON · WAVE 2 · TRACK 1 · DEFI YIELD VAULT

mbLP: deposit BTC, hold one token, earn MEZO as BTC/MUSD liquidity

Wave 1 scored 8.6 and tied first. For Wave 2 I attacked the vault against MUSD's real rules, fixed what broke, and made every claim checkable on chain.

Live on Mezo testnet | vault v6 · `0x1309…E0C0` | source verified | no owner, no admin key

mblp.surge.sh | github.com/GuTS805/Akindo

WHAT IT DOES

One BTC deposit does what takes five manual steps

Deposit BTC
one transaction, withdraw any share any time

→

MUSD trove
~75% as collateral at a 300% target
BorrowerOperations, TroveManager, HintHelpers, SortedTroves

→

Borrow MUSD
paired with the rest of the BTC

→

BTC/MUSD pool
Mezo Router, volatile pool

→

Stake in the gauge
Mezo's live gauge pays MEZO, credited per share

- **Mezo's PriceFeed** prices every share and guards every pool action with an oracle band.
- Part of each fee is posted as a **gauge incentive** (IncentiveRouter) to attract veBTC votes.
- Bitcoin wallets connect through **Mezo Passport**. Exposure stays about 1x BTC.
- Rebalancing never swaps; `rebalance()` and `harvest()` are open to anyone.

FEEDBACK TO ACTION

The Wave 1 judge asked for five things

ASK	STATUS
Handle MUSD's `maxBorrowingCapacity`	**Done.** The vault refinances its trove when a deposit would pass the cap. A 13x deposit deploys in full on testnet and mainnet forks.
A degraded exit for Recovery Mode	**Done.** Exits pay each holder their NAV share from idle assets and LP, repay debt with the LP's MUSD, leave collateral alone.
Fix the debt sum in `_snapshot`, `_troveCR`, `_netDebt`	**Done.** Pending interest was counted twice after a redistribution; the test failed before the fix.
Verify the v3 vault source	**Done.** v3, v4, v5, v6 and the IncentiveRouter are verified on the explorer.
Show real third-party usage	**Started.** The first outside deposit is on v6, and a live Community page lists every deposit from outside the team, read from chain.

THEN I ATTACKED IT

Five more gaps found by attacking my own vault

GAP	FIX, AND THE TEST THAT FAILED FIRST
Deployment sandwich: push the pool to the band edge, let a deposit or keeper step add liquidity, push back	One step adds at most a tenth of the pool. `Sandwich.t.sol`: every sandwich loses money at 2%, 6% and 10% bands.
A fall with the pool left behind: `rebalance()` reverted while the trove sat under its minimum ratio	Rebalance may cut debt outside the band. `CrashExit.t.sol`: oracle −45%, trove back above 250%.
A small deposit paid the borrowing fee for work the vault already owed	A deposit deploys only its own BTC (found by the fuzz test).
A deployment step could revert after a redemption or liquidation	New debt is capped by the collateral ratio it leaves. The old regression sequence no longer reverts.
`minBtcOut` floors only the BTC of an exit, which can be paid partly in MUSD	`withdrawMinValue` floors BTC plus MUSD at the oracle price; the app uses it.

ECONOMIC ATTACKS, MEASURED

Attackers pay to try

The attacker gets their own capital, pushes the pool to the edge of the oracle band around someone else's transaction, then pushes it back. Profit in sats (1 BTC = 100,000,000), least negative of up or down.

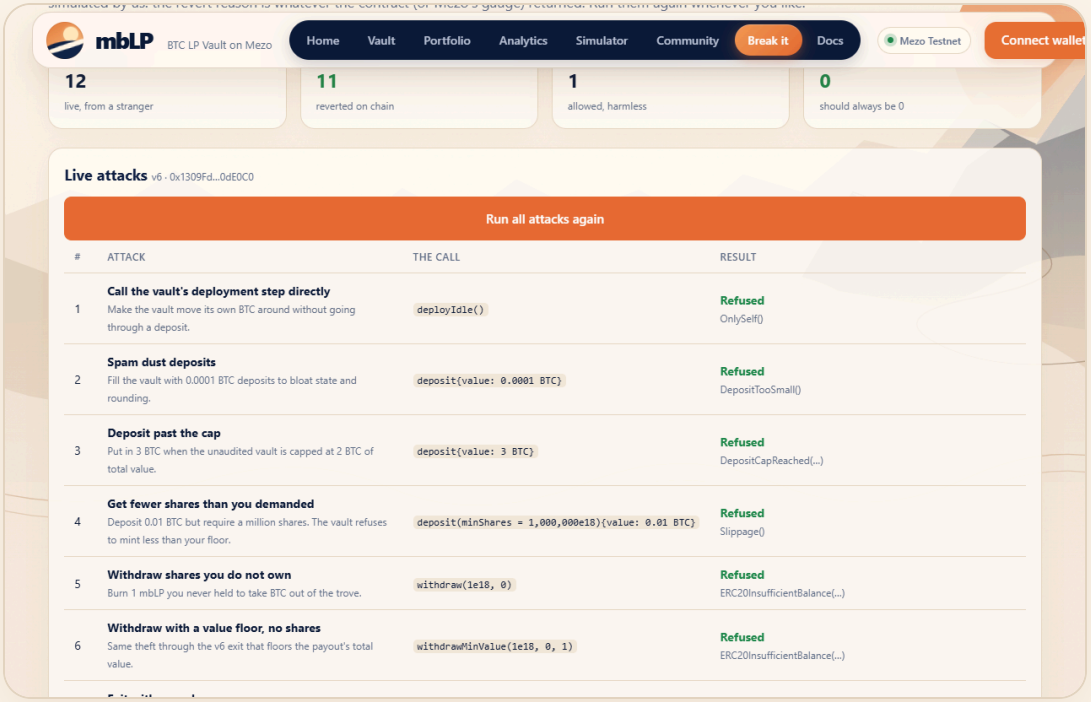
ATTACK	2% BAND (MAINNET)	6% BAND	10% BAND (TESTNET)
Sandwich a keeper step	**−53,035**	**−98,271**	**−52,019**
Sandwich a deposit	**−63,699**	**−185,763**	**−301,248**
Sandwich a withdrawal	**−66,948**	**−222,239**	**−403,379**

The first version lost here: about **+0.002 BTC per 20 BTC deployed** at a 6% band. A step cap of a fifth of the pool still lost at 8% and 10%, so the cap is a tenth. Reproduce: `forge test --match-contract SandwichTest -vv`

TRY TO BREAK IT · MBLP.SURGE.SH/BREAK

Twelve live attacks, read the revert reasons yourself

- Each row is a real call to the deployed v6 vault from a stranger's address holding 10 test BTC.
- 11 refused on chain (`OnlySelf`, `DepositCapReached`, `ERC20InsufficientBalance`, `NotAuthorized` from Mezo's gauge…).
- 1 ran and gained nothing: a stranger's `claimRewards()` returns 0.
- **0 got through.** No wallet needed; run them again any time.



LIVE ON MEZO TESTNET

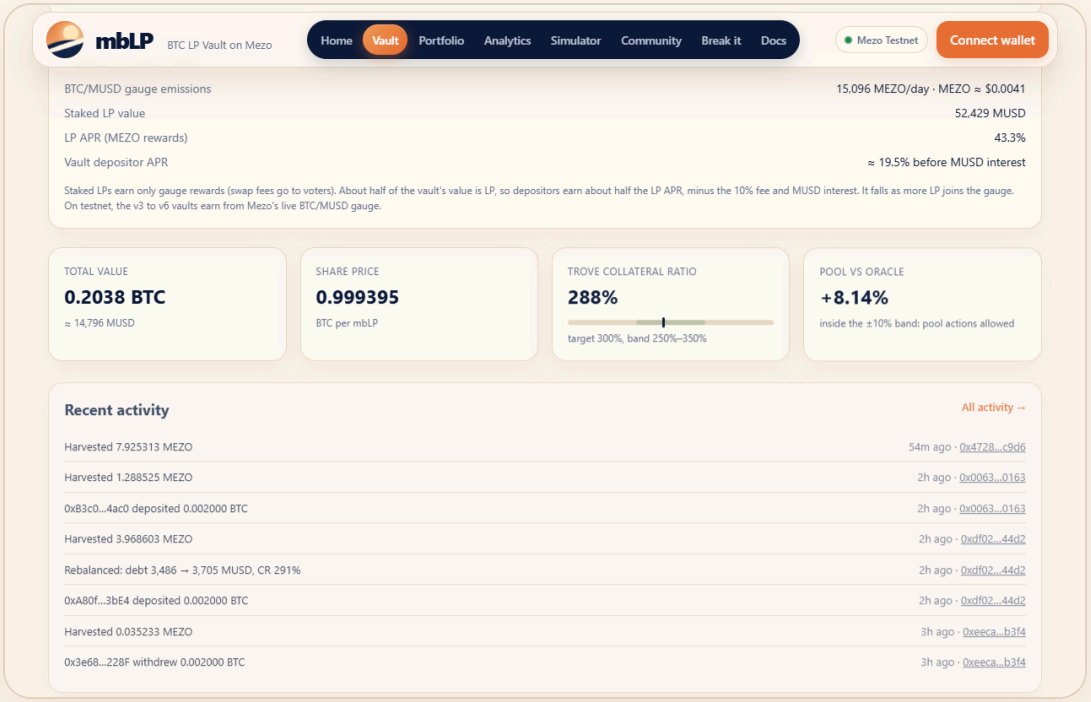
Vault v6, seeded and earning

~290%
trove collateral ratio on 12 Nov, target 300%

0.002
BTC idle: an outside deposit under the 1% rule, waiting for the next

~147
MEZO a day, measured on chain (testnet MEZO, no value)

30 min
keeper rebalances and harvests



- One deposit opened the trove, borrowed 3,521 MUSD and staked the LP in Mezo's live gauge.
- Source verified; 10% oracle band on testnet because the pool drifts 4-8%. Mainnet config: 2%.

RIGOR

Every claim has a test you can run

142
tests in 21 suites, 0 failing; 81% branch coverage

65,536
random actions in the invariant suite

5 + 3
fork tests: Mezo testnet and mainnet

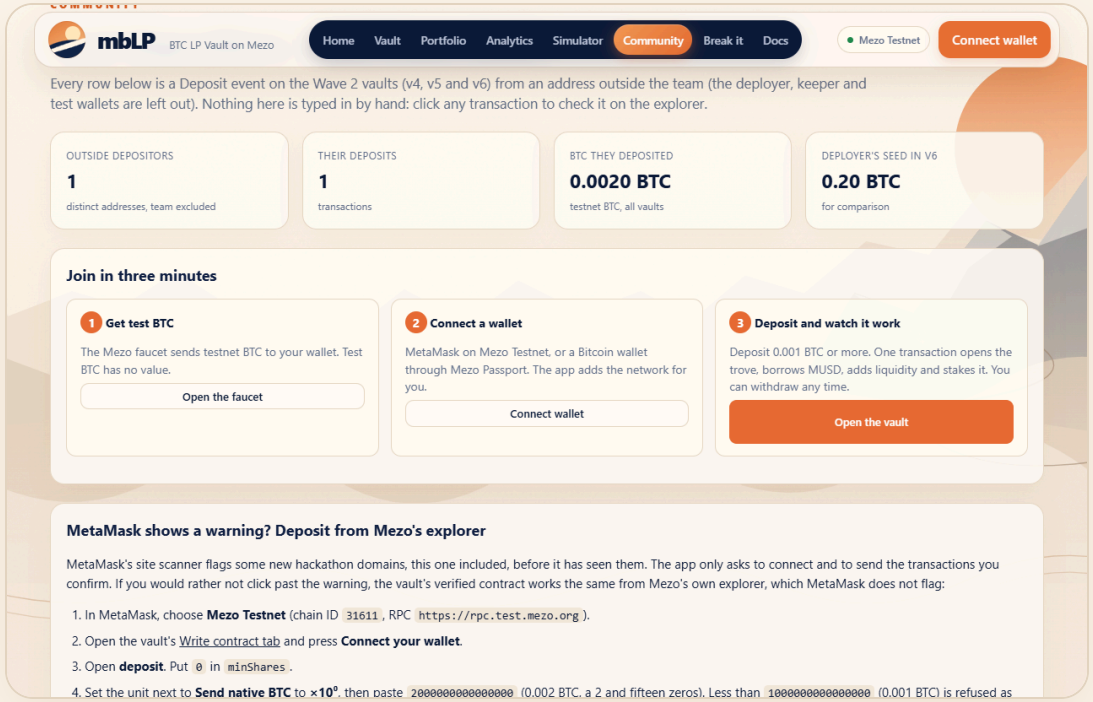
48
Slither results, each reviewed

- Fork suites run against the real MUSD, pool and gauge: a 13x deposit deploys in full and exits at (almost) full value. The main suites also run with BTC as the pool's first token, as on mainnet.
- Every fix above has a test that failed before it. The findings table is in `contracts/SECURITY.md`.
- A guarded mainnet deploy script exists (address checks, dry run, deposit cap required). It has not been run.

REAL USERS · MBLP.SURGE.SH/COMMUNITY

Every outside depositor, read from chain

- The Community page lists each Deposit event from an address outside the team (deployer, keeper and test wallets are excluded), with a link to the transaction.
- The first outside deposit: 0.002 BTC into v6, tx `0x00634449…0163`.
- Joining takes three minutes: faucet, connect, deposit 0.001 BTC or more, from the app or from Mezo's explorer.
- One asset, one transaction, withdraw any time, including in MUSD's Recovery Mode.



HONEST LIMITS AND NEXT

Testnet only, unaudited, and clear about it

Limits

- Unaudited. No owner also means no pause.
- Trusts Mezo's price feed; a stale oracle pauses deposits and trove exits.
- A thin pool can delay the last exit. In Recovery Mode about a quarter of the NAV beyond idle assets can leave at once.

Next

- Real depositors and their feedback.
- Third-party audit, then a capped mainnet launch.
- A range-managed strategy for the concentrated BTC/MUSD pool, whose mainnet gauge paid ~5x more MEZO per dollar of liquidity on 12 Nov.

Business model: 10% of the MEZO the vault earns, no deposit or exit fee. Half is posted as a gauge incentive for veBTC voters; half goes to the treasury, which pays the keeper (~$0.16 a call at mainnet gas). Anyone can call rebalance and harvest.